GenAI Using Ollama – Complete Roadmap
Goal: Take you from Beginner → Advanced GenAI Engineer using Ollama
Focus: Strong fundamentals + Real projects + Deployment-level skills
Approach: Step-by-step, industry-standard, practical implementation

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PHASE 1: FOUNDATIONS (Core Understanding of Ollama)

MODULE 1 – Ollama Fundamentals (Build Strong Base)

Objective:
Understand what Ollama is, why it exists, and how it works internally.

You Will Learn:
1. What is Ollama?
   - Local LLM runtime
   - Why local inference matters (privacy, cost, control)

2. Ollama in AI Ecosystem
   - Compare with OpenAI API
   - Compare with LM Studio
   - Compare with HuggingFace Transformers

3. Key Features
   - Local model execution
   - REST API server
   - Model management
   - Modelfile system

4. Internal Working
   - Where models are stored (~/.ollama/models)
   - Manifest files
   - Blob storage
   - How model is loaded into memory
   - CPU vs GPU usage

5. Installation (Mac / Linux / Windows)
   - Install
   - Start server
   - Pull model
   - Run first prompt

Hands-on:
- Install Ollama
- Run `ollama pull llama3`
- Run `ollama run llama3`
- Inspect model storage directory

Outcome:
You understand how Ollama works under the hood.

MODULE 2 – CLI & Model Mastery

Objective:
Master Ollama through terminal like a real engineer.

You Will Learn:

1. Model Architecture & Storage
   - Manifest
   - Blobs
   - Config files
   - Model layers

2. CLI Commands Deep Dive
   - ollama run
   - ollama pull
   - ollama list
   - ollama show
   - ollama rm
   - ollama create

3. Model Types
   - Chat models

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   - Embedding models
   - Multimodal models

4. Parameter Tuning
   - temperature
   - top_p
   - top_k
   - num_ctx
   - num_predict

5. Running Multiple Models
   - Memory management
   - Switching models
   - Performance considerations

6. Multimodal Basics
   - Image input models

Hands-on:
- Run multiple models
- Experiment with temperature differences
- Analyze performance changes

Outcome:
You can control models precisely from CLI.

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PHASE 2: PROGRAMMATIC CONTROL
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MODULE 3 – Python Integration + Modelfile

Objective:
Integrate Ollama into real applications.

You Will Learn:

1. Ollama Python Library
   - generate()
   - chat()
   - streaming responses

2. Streaming vs Non-streaming handling

3. Core Commands via Python
   - list()
   - pull()
   - show()

4. Modelfile Concept
   - FROM
   - PARAMETER
   - SYSTEM
   - TEMPLATE

5. Building Custom Models
   - Create Modelfile
   - Build model
   - Run custom system prompts

Hands-on:
- Create custom assistant model
- Build domain-specific assistant

Outcome:
You can build customized LLMs locally.

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MODULE 4 – REST API & Tool Calling
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Objective:
Understand Ollama backend like a systems engineer.

You Will Learn:

1. Ollama Server Architecture
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- Local REST server
   - Request flow

2. REST Endpoints
   - /api/generate
   - /api/chat
   - /api/embeddings
   - /api/tags

3. Using Python requests module

4. Tool Calling Basics
   - Function calling
   - Structured JSON outputs

Hands-on:
- Build small REST client
- Implement tool-calling example

Outcome:
You can integrate Ollama into backend systems.

PHASE 3: RAG (Retrieval-Augmented Generation)

MODULE 5 – RAG Stack with Ollama

Objective:
Build intelligent document-aware AI systems.

You Will Learn:

1. Embeddings Fundamentals
   - What is embedding?
   - Vector similarity
   - Cosine similarity

2. RAG Architecture
   - Chunking
   - Embedding
   - Vector DB
   - Retrieval
   - Generation

3. /api/embeddings endpoint

4. Vector Databases
   - FAISS
   - ChromaDB
   - Weaviate

5. Data Ingestion
   - PDFs
   - CSVs
   - Text files

6. LangChain Integration

Hands-on Project (Minor Project):
Build Local RAG App:
- Upload PDFs
- Store embeddings
- Ask questions privately

Outcome:
You can build ChatGPT-like system over your own documents.

PHASE 4: PRO-LEVEL DEVELOPMENT

MODULE 6 – Advanced Performance & Deployment

Objective:
Operate Ollama like production engineer.
You Will Learn:

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1. Running Multiple Models Concurrently
2. Model Chaining (Pipeline design)
3. GPU Offloading
4. System Monitoring
5. Ollama Cloud
6. HuggingFace Model Import
7. Ollama Desktop App

Hands-on:
- Build model pipeline
- Benchmark performance
- Optimize runtime

Outcome:
You understand scaling & efficiency.

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PHASE 5: CAPSTONE PROJECT
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MODULE 7 – AI Brand Monitoring System

Objective:
Build Full-Stack AI System (Production Ready)

Project Components:

1. Data Collection
   - Reddit API integration
   - Store brand mentions

2. Database Design
   - PostgreSQL / SQLite schema

3. Sentiment Analysis using Ollama

4. Context Understanding

5. Dashboard
   - Python (Streamlit / Dash)
   - Visual charts

6. Dockerization
   - Dockerfile
   - docker-compose
   - Container networking

7. Deployment-ready architecture

Outcome:
You build real-world AI product.

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PHASE 6: ALTERNATIVE ECOSYSTEM
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MODULE 8 – LM Studio Comparison

Objective:
Understand alternative local LLM runtime.

You Will Learn:

1. LM Studio installation
2. Running models
3. Model tuning
4. Multimodal input
5. Developer tab
6. Python integration
7. Comparison with Ollama
8. HuggingFace integration

Outcome:
You understand trade-offs between tools.

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FINAL SKILLSET YOU WILL HAVE
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✓ Deep Ollama Fundamentals
✓ CLI Mastery
✓ Python Integration
✓ REST API Engineering
✓ Tool Calling
✓ RAG Architecture
✓ Vector Databases
✓ Performance Optimization
✓ Multi-model orchestration
✓ Dockerized AI Deployment
✓ Full-stack AI system building

RECOMMENDED LEARNING STRATEGY

Week 1–2: Modules 1–2
Week 3–4: Modules 3–4
Week 5–6: Module 5 (RAG)
Week 7: Module 6
Week 8–9: Capstone Project
Week 10: LM Studio Comparison + Optimization

END RESULT

You move from:
"Running models locally"

to:

"Designing, building, optimizing, and deploying production-grade
AI systems powered by Ollama."